

Batch emissions report

exp2_rag_2026-10-10_12-19-41

Models: granite4:3b-h

Question ordering: Keep original order

Shuffle seed: None

Pipeline: RAG | Questions: 4 | Successful: 4 | Failed/unknown: 0

Batch runtime: 247.49 s | Successful requests/min: 0.96974

Measured cost (all requests)	Observed sum	Valid / all
Energy (Wh)	1.9681	4 / 4
Emissions (g CO2e)	1.4041	4 / 4

Method and data quality

Statistics and distributions use successful requests with finite measurements. Cost totals include measured failed requests. Missing/non-finite values are excluded, never replaced with zero. Partial totals are not estimates of unmeasured cost.

Variance and standard deviation use the sample definition ($n - 1$); both are N/A for fewer than two observations. Variance has squared units. P95 uses linear interpolation. Small values use significant digits or scientific notation.

Historical pipelines may already have stored missing telemetry as zero. This report cannot distinguish those defaults from measured zero. RAM fields describe estimated energy, not memory usage. Per-question charts use CSV row order; red points mark failed/unknown requests.

Metric	Missing	Zero
Latency (s)	0	0
Energy (Wh)	0	0
Emissions (g CO2e)	0	0
Retrieval latency (s)	0	0
Retrieval energy (Wh)	0	0
Retrieval emissions (g CO2e)	0	0
Generation latency (s)	0	0
Generation energy (Wh)	0	0
Generation emissions (g CO2e)	0	0
CPU energy (Wh)	0	0
GPU energy (Wh)	0	4
RAM energy (Wh)	0	0

Measurement scope and reliability

Machine scope includes supported CPU/GPU/RAM activity from local Ollama and other processes; it does not isolate model energy or measure wall-socket whole-system power. Unknown backends are unverified; RAM is an energy estimate, not memory usage. A local endpoint does not prove all accelerator hardware is supported.

Tracking sessions recorded: 8. Expected for this CSV: 8. Metadata describes configuration and backend evidence, not independent meter validation.

codecarbon_version: 3.3.0

os: Windows-11-10.0.26200-SP0

effective_scope: machine

measure_power_secs: 1

pue: 1.0

country_iso_code: IND

Config Profile: "inference"

Configured Settings: {"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}

CPU backend: unknown / not exposed by this version

GPU backend: unknown / not exposed by this version

RAM backend: unknown / not exposed by this version

Hardware: {'cpu_model': '12th Gen Intel(R) Core(TM) i5-1235U', 'gpu_model': '', 'gpu_count': 0, 'ram_total_size': 15.69796371459961}

Backend objects/modes: [{'class': 'RAM', 'mode': 'unknown'}, {'class': 'CPU', 'mode': 'windows_emi'}]

Sensor-backed versus estimated: backend names containing constant, TDP or CPU load indicate estimation; RAPL, NVML and power metrics indicate hardware telemetry. Unknown modes remain unverified. Emissions are calculated estimates in all cases.

Tracker stop failures: 0

A partial observed sum excludes unavailable readings. A complete batch cost cannot be claimed when coverage is incomplete. Zero readings are retained; unknown hardware coverage must not be interpreted as zero consumption.

Statistical summary

Latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	58.189	70.57	817.85	28.598	15.786	75.83	75.693

Energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.49202	0.59673	0.05683	0.23839	0.13789	0.63671	0.63572

Emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.35102	0.42573	0.028927	0.17008	0.098378	0.45426	0.45355

Retrieval latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.084854	0.048694	0.0067122	0.081928	0.034728	0.2073	0.18394

Retrieval energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.00061973	0.00029826	7.7634e-07	0.0008811	6.1861e-06	0.0018762	0.0016829

Retrieval emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.00044214	0.00021279	3.9516e-07	0.00062861	4.4134e-06	0.0013386	0.0012006

Generation latency (s)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	58.104	70.443	817.29	28.588	15.734	75.796	75.656

Generation energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.4914	0.59579	0.056879	0.23849	0.13731	0.63671	0.63571

Generation emissions (g CO2e)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.35058	0.42506	0.028951	0.17015	0.097959	0.45425	0.45354

CPU energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.33053	0.40087	0.025298	0.15905	0.094072	0.42631	0.4257

GPU energy (Wh)

Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0	0	0	0	0	0	0

RAM energy (Wh)

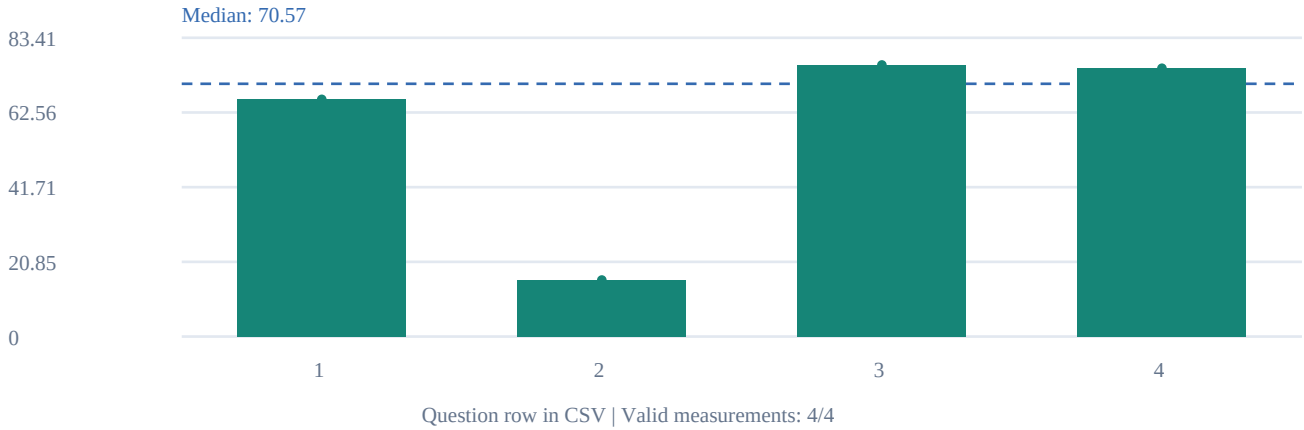
Valid n	Mean	Median	Variance	Std dev	Min	Max	P95
4	0.16148	0.19585	0.0062959	0.079346	0.043821	0.21041	0.21002

Which questions cost the most?

Bars show individual requests; large batches use points to avoid overcrowding. Missing measurements are marked X, not shown as zero.

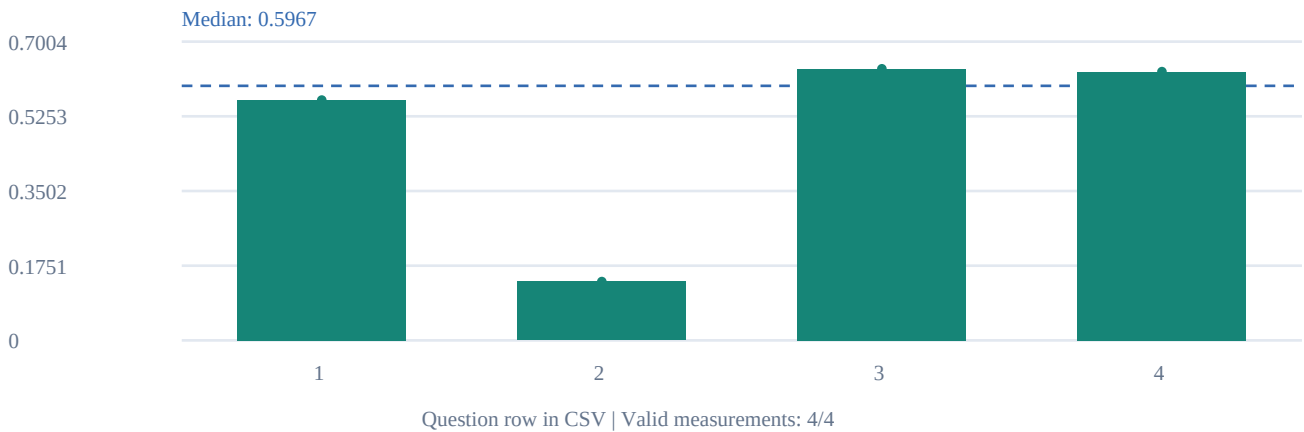
Latency (s)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median



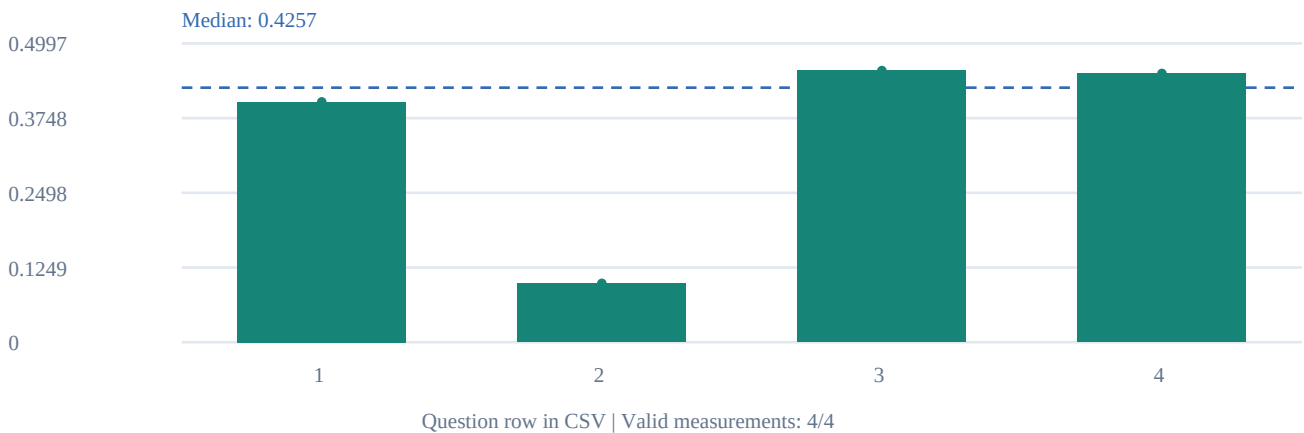
Energy (Wh)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median



Emissions (g CO2e)

Teal: successful | Red: failed/unknown | X: missing | Blue line: successful median

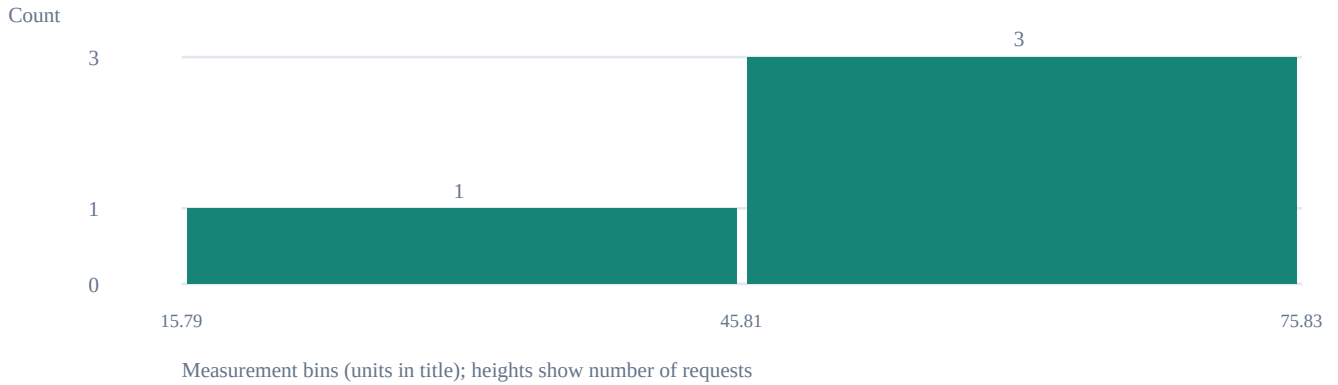


How much do requests vary?

Successful requests only. Bin boundaries show measurement values; bar heights and labels show request counts.

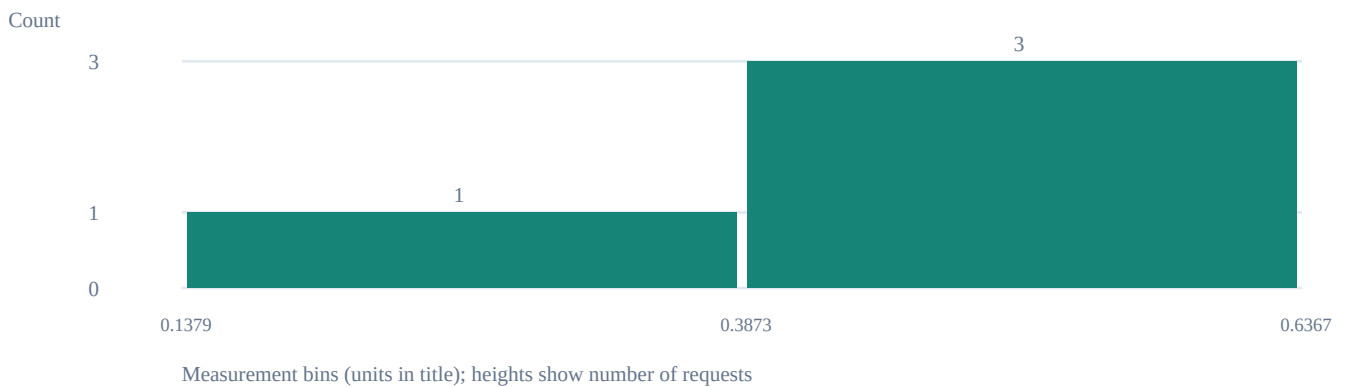
Latency (s)

Successful requests: n=4 | Mean 58.19 | Median 70.57



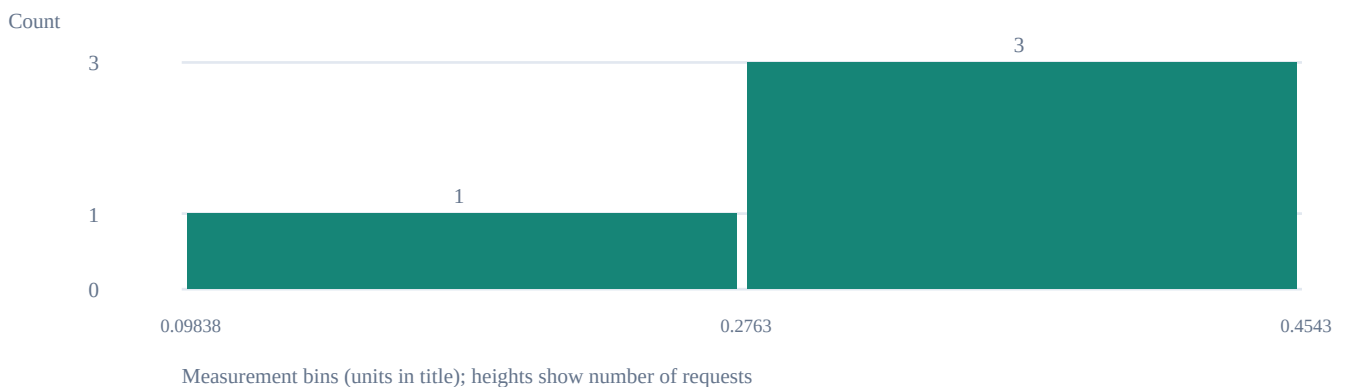
Energy (Wh)

Successful requests: n=4 | Mean 0.492 | Median 0.5967



Emissions (g CO2e)

Successful requests: n=4 | Mean 0.351 | Median 0.4257



Where is energy spent?

Compare observed component costs below. Amber bars indicate incomplete coverage; comparisons with different coverage may be misleading. These are tracking estimates, not isolated model or wall-socket measurements.

Hardware energy (Wh)

Observed sums across all requests; partial coverage is labelled.



Retrieval versus generation

Time (s)

Observed sums across all requests; partial coverage is labelled.



Energy (Wh)

Observed sums across all requests; partial coverage is labelled.



Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.



Question execution mapping

Chart question rows correspond to this execution sequence. Original serial numbers are local to each topic sheet.

Position	Topic sheet	Serial	Question type
1	Customer Operations	13	Follow-Up 1
2	Customer Operations	14	Follow-Up 2
3	Customer Operations	15	Follow-Up 3
4	Customer Operations	16	Follow-Up 4

RAG evidence and fallback overview

Evidence-backed	Base-model fallback	Not recorded
3	1	0

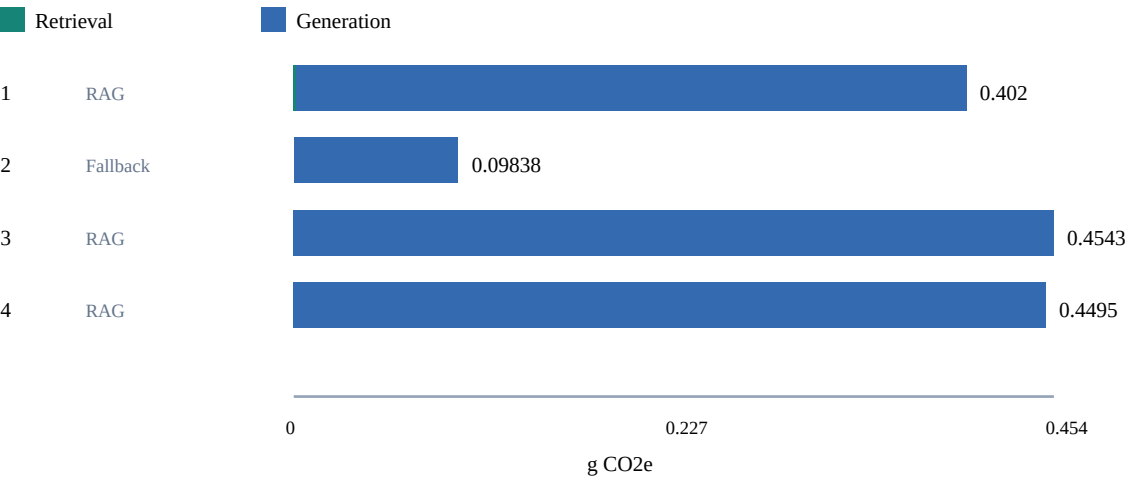
Counts cover attempted question rows, including failed or cancelled attempts. Evidence acceptance is a retrieval heuristic, not an answer-quality score. Unknown decisions are never inferred from historical chunks.

Fallback reason	Questions
insufficient evidence	1

Per-question retrieval and generation emissions

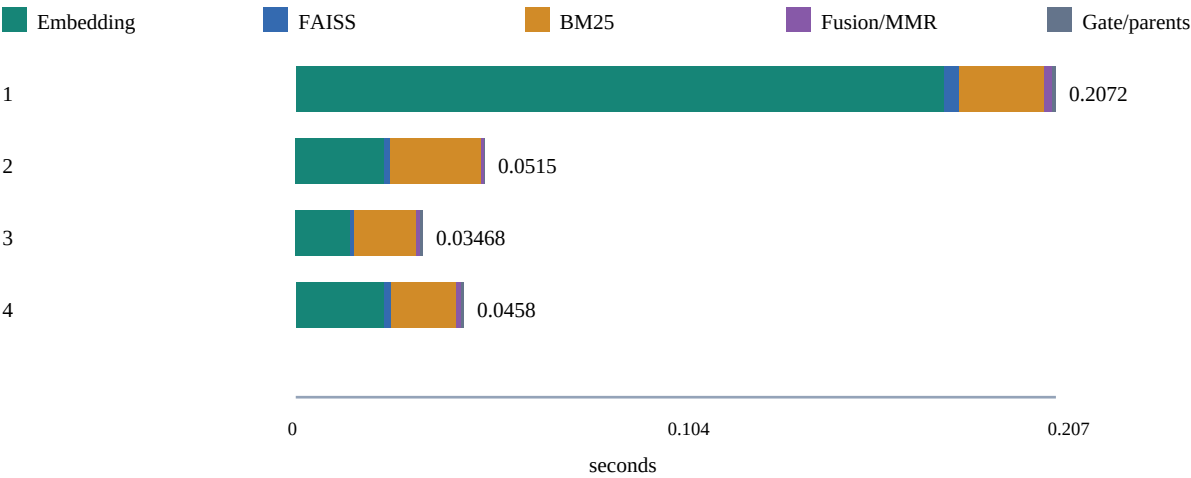
Horizontal stacks show retrieval plus generation for each execution position. Failed and cancelled attempts may include costs; missing phase readings are marked N/A. Corpus costs are excluded.

Emissions by execution position



Retrieval-step timings

Retrieval suboperation time



These steps share one retrieval emissions measurement. Timer sums need not equal tracker duration because setup, context formatting and other overhead are excluded.

Evidence selection by question

Positions follow execution order. Child candidates are expanded into unique parent sources. Source details refer to supplied evidence, not verified citations in the generated answer.

Position	Mode / status	Children / parents	Best cosine	Gate reason / source files
1	Evidence-backed / ok	4 / 4	0.25973	retriever agreement nlpcarbon.txt, mindweb.txt, green-automated-ml.txt, chasingcarbon.txt
2	Fallback / ok	4 / 0	0.19621	insufficient evidence No sources recorded
3	Evidence-backed / ok	4 / 4	0.34873	dense similarity mindweb.txt, paris_agreement.txt, trilemma.txt
4	Evidence-backed / ok	4 / 4	0.30989	dense similarity paris_agreement.txt, mindweb.txt

Evidence-backed versus fallback costs

Successful requests with recorded decisions only. Each metric uses available finite readings and shows valid n. Different question difficulty can explain cost differences; this is not a causal efficiency ranking.

Response mode	Successful n	Mean emissions (g) / n	Mean energy (Wh) / n	Mean latency (s) / n
Evidence-backed	3	0.43524 / 3	0.61006 / 3	72.323 / 3
Fallback	1	0.098378 / 1	0.13789 / 1	15.786 / 1

Recorded retrieval configuration

Setting	Value
Index generation	56d964e846804beabae81034205e793a
Embedding Model	all-MiniLM-L6-v2
Index Type	IndexFlatIP
Normalize Embeddings	True
Dense Candidates	15
Sparse Candidates	15
Rrf Constant	60
Selected Children	4
Mmr Lambda	0.75
Evidence Similarity Threshold	0.3
Context Characters	12000
Child Target Tokens	180
Parent Target Tokens	900
Sentence Overlap	1

Settings are the run snapshot, not the current index configuration. The context budget covers evidence text only. Corpus preparation costs remain in the indexing report and are not added to these question totals.